VP07 — Features across the architecture

Revision: `e2a2f39e5c78476bb1bba08e406114367eb59953f02a63e952e435c880d24007`.

contributes-to, owns and explicit allocated-to per mode. Unspecified allocation is reported as a gap.

Complete logical architecture and feature realization — mode NormalDesktop

```
Mermaid: Label construct outside XFMD Flowchart 1
flowchart LR
    n_Application["Application (unit)"]
    n_ApplicationPreferences(["ApplicationPreferences
(functionality)"])
    n_ChapterNavigation{{"ChapterNavigation (feature)"}}
    n_ConfigurableScrolling{{"ConfigurableScrolling (feature)"}}
    n_DiagramLayout(["DiagramLayout (functionality)"])
    n_DiagramPreparation(["DiagramPreparation (functionality)"])
    n_DocumentIndex(["DocumentIndex (functionality)"])
    n_DocumentIndexService["DocumentIndexService (unit)"]
    n_DocumentNavigation{{"DocumentNavigation (feature)"}}
    n_DocumentReferences{{"DocumentReferences (feature)"}}
    n_DocumentSession(["DocumentSession (functionality)"])
    n_DocumentStorage["DocumentStorage (unit)"]
    n_Documents["Documents (unit)"]
    n_EmbeddedVisuals(["EmbeddedVisuals (functionality)"])
    n_EventLoop["EventLoop (unit)"]
    n_EventScheduling(["EventScheduling (functionality)"])
    n_ExternalLauncher["ExternalLauncher (unit)"]
    n_ExternalViewBroker["ExternalViewBroker (unit)"]
    n_ExternalViewGenerator["ExternalViewGenerator (unit)"]
    n_FilteredFileTree(["FilteredFileTree (functionality)"])
```

```
    n_FoxPresentation(["FoxPresentation (functionality)"])
    n_GeneratedViews["GeneratedViews (unit)"]
    n_Identity["Identity (unit)"]
    n_Interpreter["Interpreter (unit)"]
    n_LivePreview{{"LivePreview (feature)"}}
    n_LocalFileStorage(["LocalFileStorage (functionality)"])
    n_MarkdownInterpretation(["MarkdownInterpretation
(functionality)"])
    n_MarkdownPresentation{{"MarkdownPresentation (feature)"}}
    n_MermaidDiagrams{{"MermaidDiagrams (feature)"}}
    n_MermaidInterpretation(["MermaidInterpretation
(functionality)"])
    n_NativeShell["NativeShell (unit)"]
    n_Navigation["Navigation (unit)"]
    n_NavigationHistory(["NavigationHistory (functionality)"])
    n_PageComposition(["PageComposition (functionality)"])
    n_PagedPublication{{"PagedPublication (feature)"}}
    n_PdfExport(["PdfExport (functionality)"])
    n_Preferences["Preferences (unit)"]
    n_Preview["Preview (unit)"]
    n_PreviewPipeline(["PreviewPipeline (functionality)"])
    n_PreviewTextSelection(["PreviewTextSelection (functionality)"])
    n_Publication["Publication (unit)"]
    n_RenderLayout(["RenderLayout (functionality)"])
    n_Renderer["Renderer (unit)"]
    n_ScrollMotion(["ScrollMotion (functionality)"])
    n_SdpProjects["SdpProjects (unit)"]
    n_SourceAnchorMapping(["SourceAnchorMapping (functionality)"])
    n_SynchronizedScrolling{{"SynchronizedScrolling (feature)"}}
    n_TextEditing(["TextEditing (functionality)"])
    n_Typography["Typography (unit)"]
    n_TypographyAndDrawing(["TypographyAndDrawing (functionality)"])
    n_Viewport["Viewport (unit)"]
    n_VisualResources["VisualResources (unit)"]
    n_WorkPathHistory(["WorkPathHistory (functionality)"])
    n_WorkPathNavigation{{"WorkPathNavigation (feature)"}}
    n_Workspace["Workspace (unit)"]
    n_WorkspaceControls(["WorkspaceControls (functionality)"])
```

```
    n_XfmdDesktop["XfmdDesktop (container)"]
    n_Application -->|contains| n_DocumentIndexService
    n_Application -->|contains| n_DocumentStorage
    n_Application -->|contains| n_Documents
    n_Application -->|contains| n_EventLoop
    n_Application -->|contains| n_GeneratedViews
    n_Application -->|contains| n_Identity
    n_Application -->|contains| n_NativeShell
    n_Application -->|contains| n_Navigation
    n_Application -->|contains| n_Preferences
    n_Application -->|contains| n_Preview
    n_Application -->|contains| n_Publication
    n_Application -->|contains| n_SdpProjects
    n_Application -->|contains| n_Typography
    n_Application -->|contains| n_Viewport
    n_Application -->|contains| n_VisualResources
    n_Application -->|contains| n_Workspace
    n_ApplicationPreferences -->|allocated-to| n_XfmdDesktop
    n_ApplicationPreferences -.->|contributes-to|
n_ConfigurableScrolling
    n_ApplicationPreferences -.->|contributes-to|
n_DocumentNavigation
    n_ApplicationPreferences -.->|contributes-to|
n_DocumentReferences
    n_ApplicationPreferences -.->|contributes-to| n_PagedPublication
    n_DiagramLayout -->|allocated-to| n_XfmdDesktop
    n_DiagramLayout -.->|contributes-to| n_MermaidDiagrams
    n_DiagramPreparation -->|allocated-to| n_XfmdDesktop
    n_DiagramPreparation -.->|contributes-to| n_MermaidDiagrams
    n_DocumentIndex -->|allocated-to| n_XfmdDesktop
    n_DocumentIndex -.->|contributes-to| n_ChapterNavigation
    n_DocumentIndex -.->|contributes-to| n_DocumentReferences
    n_DocumentIndexService -->|owns| n_DocumentIndex
    n_DocumentSession -->|allocated-to| n_XfmdDesktop
    n_DocumentSession -.->|contributes-to| n_DocumentNavigation
    n_DocumentSession -.->|contributes-to| n_LivePreview
    n_DocumentSession -.->|contributes-to| n_MarkdownPresentation
    n_DocumentSession -.->|contributes-to| n_PagedPublication
```

```
    n_DocumentStorage -->|owns| n_LocalFileStorage
    n_Documents -->|owns| n_DocumentSession
    n_Documents -->|owns| n_TextEditing
    n_EmbeddedVisuals -->|allocated-to| n_XfmdDesktop
    n_EmbeddedVisuals -.->|contributes-to| n_MarkdownPresentation
    n_EventLoop -->|owns| n_EventScheduling
    n_EventScheduling -->|allocated-to| n_XfmdDesktop
    n_EventScheduling -.->|contributes-to| n_LivePreview
    n_FilteredFileTree -->|allocated-to| n_XfmdDesktop
    n_FilteredFileTree -.->|contributes-to| n_WorkPathNavigation
    n_FoxPresentation -->|allocated-to| n_XfmdDesktop
    n_FoxPresentation -.->|contributes-to| n_ConfigurableScrolling
    n_FoxPresentation -.->|contributes-to| n_DocumentNavigation
    n_FoxPresentation -.->|contributes-to| n_LivePreview
    n_FoxPresentation -.->|contributes-to| n_MarkdownPresentation
    n_FoxPresentation -.->|contributes-to| n_MermaidDiagrams
    n_FoxPresentation -.->|contributes-to| n_PagedPublication
    n_FoxPresentation -.->|contributes-to| n_SynchronizedScrolling
    n_Interpreter -->|owns| n_MarkdownInterpretation
    n_Interpreter -->|owns| n_MermaidInterpretation
    n_LocalFileStorage -->|allocated-to| n_XfmdDesktop
    n_LocalFileStorage -.->|contributes-to| n_DocumentNavigation
    n_LocalFileStorage -.->|contributes-to| n_MarkdownPresentation
    n_MarkdownInterpretation -->|allocated-to| n_XfmdDesktop
    n_MarkdownInterpretation -.->|contributes-to| n_LivePreview
    n_MarkdownInterpretation -.->|contributes-to|
n_MarkdownPresentation
    n_MarkdownInterpretation -.->|contributes-to| n_MermaidDiagrams
    n_MarkdownInterpretation -.->|contributes-to| n_PagedPublication
    n_MermaidInterpretation -->|allocated-to| n_XfmdDesktop
    n_MermaidInterpretation -.->|contributes-to| n_MermaidDiagrams
    n_NativeShell -->|owns| n_FoxPresentation
    n_NativeShell -->|owns| n_WorkspaceControls
    n_Navigation -->|owns| n_NavigationHistory
    n_NavigationHistory -->|allocated-to| n_XfmdDesktop
    n_NavigationHistory -.->|contributes-to| n_ChapterNavigation
    n_NavigationHistory -.->|contributes-to| n_DocumentNavigation
    n_NavigationHistory -.->|contributes-to| n_DocumentReferences
```

```
    n_PageComposition -->|allocated-to| n_XfmdDesktop
    n_PageComposition -.->|contributes-to| n_MarkdownPresentation
    n_PageComposition -.->|contributes-to| n_MermaidDiagrams
    n_PageComposition -.->|contributes-to| n_PagedPublication
    n_PdfExport -->|allocated-to| n_XfmdDesktop
    n_PdfExport -.->|contributes-to| n_MermaidDiagrams
    n_PdfExport -.->|contributes-to| n_PagedPublication
    n_Preferences -->|owns| n_ApplicationPreferences
    n_Preview -->|owns| n_PreviewPipeline
    n_Preview -->|owns| n_PreviewTextSelection
    n_PreviewPipeline -->|allocated-to| n_XfmdDesktop
    n_PreviewPipeline -.->|contributes-to| n_LivePreview
    n_PreviewPipeline -.->|contributes-to| n_MarkdownPresentation
    n_PreviewPipeline -.->|contributes-to| n_MermaidDiagrams
    n_PreviewPipeline -.->|contributes-to| n_PagedPublication
    n_PreviewTextSelection -->|allocated-to| n_XfmdDesktop
    n_PreviewTextSelection -.->|contributes-to|
n_MarkdownPresentation
    n_PreviewTextSelection -.->|contributes-to| n_MermaidDiagrams
    n_Publication -->|owns| n_PdfExport
    n_RenderLayout -->|allocated-to| n_XfmdDesktop
    n_RenderLayout -.->|contributes-to| n_LivePreview
    n_RenderLayout -.->|contributes-to| n_MarkdownPresentation
    n_RenderLayout -.->|contributes-to| n_MermaidDiagrams
    n_RenderLayout -.->|contributes-to| n_PagedPublication
    n_RenderLayout -.->|contributes-to| n_SynchronizedScrolling
    n_Renderer -->|owns| n_DiagramLayout
    n_Renderer -->|owns| n_PageComposition
    n_Renderer -->|owns| n_RenderLayout
    n_ScrollMotion -->|allocated-to| n_XfmdDesktop
    n_ScrollMotion -.->|contributes-to| n_ConfigurableScrolling
    n_ScrollMotion -.->|contributes-to| n_SynchronizedScrolling
    n_SourceAnchorMapping -->|allocated-to| n_XfmdDesktop
    n_SourceAnchorMapping -.->|contributes-to| n_ChapterNavigation
    n_SourceAnchorMapping -.->|contributes-to|
n_ConfigurableScrolling
    n_SourceAnchorMapping -.->|contributes-to| n_DocumentNavigation
    n_SourceAnchorMapping -.->|contributes-to| n_DocumentReferences
```

```
    n_SourceAnchorMapping -.->|contributes-to| n_PagedPublication
    n_SourceAnchorMapping -.->|contributes-to|
n_SynchronizedScrolling
    n_TextEditing -->|allocated-to| n_XfmdDesktop
    n_TextEditing -.->|contributes-to| n_ConfigurableScrolling
    n_TextEditing -.->|contributes-to| n_LivePreview
    n_Typography -->|owns| n_TypographyAndDrawing
    n_TypographyAndDrawing -->|allocated-to| n_XfmdDesktop
    n_TypographyAndDrawing -.->|contributes-to|
n_MarkdownPresentation
    n_TypographyAndDrawing -.->|contributes-to| n_MermaidDiagrams
    n_TypographyAndDrawing -.->|contributes-to| n_PagedPublication
    n_Viewport -->|owns| n_ScrollMotion
    n_Viewport -->|owns| n_SourceAnchorMapping
    n_VisualResources -->|owns| n_DiagramPreparation
    n_VisualResources -->|owns| n_EmbeddedVisuals
    n_WorkPathHistory -->|allocated-to| n_XfmdDesktop
    n_WorkPathHistory -.->|contributes-to| n_WorkPathNavigation
    n_Workspace -->|owns| n_FilteredFileTree
    n_Workspace -->|owns| n_WorkPathHistory
    n_WorkspaceControls -->|allocated-to| n_XfmdDesktop
    n_WorkspaceControls -.->|contributes-to| n_ChapterNavigation
    n_WorkspaceControls -.->|contributes-to| n_ConfigurableScrolling
    n_WorkspaceControls -.->|contributes-to| n_DocumentNavigation
    n_WorkspaceControls -.->|contributes-to| n_DocumentReferences
    n_WorkspaceControls -.->|contributes-to| n_MarkdownPresentation
    n_WorkspaceControls -.->|contributes-to| n_PagedPublication
    n_WorkspaceControls -.->|contributes-to| n_SynchronizedScrolling
    n_WorkspaceControls -.->|contributes-to| n_WorkPathNavigation
    n_XfmdDesktop -->|contains| n_Application
    n_XfmdDesktop -->|contains| n_Interpreter
    n_XfmdDesktop -->|contains| n_Renderer
```

Source facts: f0019, f0020, f0021, f0022, f0023, f0024, f0025, f0026, f0027, f0028, f0029, f0030, f0031, f0032, f0033, f0034, f0046, f0047, f0049, f0050, f0051, f0129, f0130, f0135, f0136, f0176, f0177, f0178, f0181, f0207, f0208, f0209, f0210, f0211, f0222, f0232, f0233, f0274, f0275, f0276, f0280, f0281, f0349, f0350, f0351, f0352, f0353, f0354, f0355, f0356, f0357, f0358, f0484, f0485, f0523, f0524, f0525, f0531,

f0532, f0533, f0534, f0535, f0539, f0540, f0544, f0545, f0576, f0587, f0588, f0589, f0590, f0600, f0601, f0602, f0603, f0625, f0626, f0627, f0652, f0687, f0688, f0717, f0718, f0719, f0720, f0721, f0740, f0741, f0742, f0747, f0837, f0838, f0839, f0840, f0841, f0842, f0844, f0845, f0846, f0889, f0890, f0891, f0907, f0908, f0909, f0910, f0911, f0912, f0913, f0936, f0937, f0938, f0944, f0946, f0947, f0948, f0949, f0965, f0966, f0983, f0984, f0996, f0997, f0999, f1000, f1013, f1014, f1015, f1016, f1017, f1018, f1019, f1020, f1022, f1031, f1032, f1033.

Scope and notation

All declared Systems, containers and units form the logical architecture base, including isolated declarations. Hexagons are Features; rounded nodes are contributing Functionality. Dotted contributes-to edges overlay feature realization on explicit contains and owns relationships. This is decomposition and responsibility, not an inferred deployment or call graph.

Selected mode: **NormalDesktop**. Only this mode's explicit allocated-to edges are included. Ownership does not imply allocation; missing placement is reported below.

Feature contributions

Feature	Functionality	Logical owner	Allocation in selected scope	Contribution fact
ChapterNavigation	DocumentIndex	DocumentIndexService	XfmdDesktop	f0177
ChapterNavigation	NavigationHistory	Navigation	XfmdDesktop	f0588
ChapterNavigation	SourceAnchorMapping	Viewport	XfmdDesktop	f0908
ChapterNavigation	WorkspaceControls	NativeShell	XfmdDesktop	f1014

ConfigurableScrolling	ApplicationPreferences	Preferences	XfmdDesktop	f0047
ConfigurableScrolling	FoxPresentation	NativeShell	XfmdDesktop	f0352
ConfigurableScrolling	ScrollMotion	Viewport	XfmdDesktop	f0890
ConfigurableScrolling	SourceAnchorMapping	Viewport	XfmdDesktop	f0909
ConfigurableScrolling	TextEditing	Documents	XfmdDesktop	f0937
ConfigurableScrolling	WorkspaceControls	NativeShell	XfmdDesktop	f1015
DocumentNavigation	ApplicationPreferences	Preferences	XfmdDesktop	f0049
DocumentNavigation	DocumentSession	Documents	XfmdDesktop	f0208
DocumentNavigation	FoxPresentation	NativeShell	XfmdDesktop	f0353
DocumentNavigation	LocalFileStorage	DocumentStorage	XfmdDesktop	f0524
DocumentNavigation	NavigationHistory	Navigation	XfmdDesktop	f0589

DocumentNavigation	SourceAnchorMapping	Viewport	XfmdDesktop	f0910
DocumentNavigation	WorkspaceControls	NativeShell	XfmdDesktop	f1016
DocumentReferences	ApplicationPreferences	Preferences	XfmdDesktop	f0050
DocumentReferences	DocumentIndex	DocumentIndexService	XfmdDesktop	f0178
DocumentReferences	NavigationHistory	Navigation	XfmdDesktop	f0590
DocumentReferences	SourceAnchorMapping	Viewport	XfmdDesktop	f0911
DocumentReferences	WorkspaceControls	NativeShell	XfmdDesktop	f1017
LivePreview	DocumentSession	Documents	XfmdDesktop	f0209
LivePreview	EventScheduling	EventLoop	XfmdDesktop	f0281
LivePreview	FoxPresentation	NativeShell	XfmdDesktop	f0354
LivePreview	MarkdownInterpretation	Interpreter	XfmdDesktop	f0532
LivePreview	PreviewPipeline	Preview	XfmdDesktop	f0718

LivePreview	RenderLayout	Renderer	XfmdDesktop	f0838
LivePreview	TextEditing	Documents	XfmdDesktop	f0938
MarkdownPresentation	DocumentSession	Documents	XfmdDesktop	f0210
MarkdownPresentation	EmbeddedVisuals	VisualResources	XfmdDesktop	f0275
MarkdownPresentation	FoxPresentation	NativeShell	XfmdDesktop	f0355
MarkdownPresentation	LocalFileStorage	DocumentStorage	XfmdDesktop	f0525
MarkdownPresentation	MarkdownInterpretation	Interpreter	XfmdDesktop	f0533
MarkdownPresentation	PageComposition	Renderer	XfmdDesktop	f0601
MarkdownPresentation	PreviewPipeline	Preview	XfmdDesktop	f0719
MarkdownPresentation	PreviewTextSelection	Preview	XfmdDesktop	f0741

MarkdownPresentation	RenderLayout	Renderer	XfmdDesktop	f0839
MarkdownPresentation	TypographyAndDrawing	Typography	XfmdDesktop	f0947
MarkdownPresentation	WorkspaceControls	NativeShell	XfmdDesktop	f1018
MermaidDiagrams	DiagramLayout	Renderer	XfmdDesktop	f0130
MermaidDiagrams	DiagramPreparation	VisualResources	XfmdDesktop	f0136
MermaidDiagrams	FoxPresentation	NativeShell	XfmdDesktop	f0356
MermaidDiagrams	MarkdownInterpretation	Interpreter	XfmdDesktop	f0534
MermaidDiagrams	MermaidInterpretation	Interpreter	XfmdDesktop	f0540
MermaidDiagrams	PageComposition	Renderer	XfmdDesktop	f0602
MermaidDiagrams	PdfExport	Publication	XfmdDesktop	f0626
MermaidDiagrams	PreviewPipeline	Preview	XfmdDesktop	f0720
MermaidDiagrams	PreviewTextSelection	Preview	XfmdDesktop	f0742
MermaidDiagrams	RenderLayout	Renderer	XfmdDesktop	f0840

MermaidDiagrams	TypographyAndDrawing	Typography	XfmdDesktop	f0948
PagedPublication	ApplicationPreferences	Preferences	XfmdDesktop	f0051
PagedPublication	DocumentSession	Documents	XfmdDesktop	f0211
PagedPublication	FoxPresentation	NativeShell	XfmdDesktop	f0357
PagedPublication	MarkdownInterpretation	Interpreter	XfmdDesktop	f0535
PagedPublication	PageComposition	Renderer	XfmdDesktop	f0603
PagedPublication	PdfExport	Publication	XfmdDesktop	f0627
PagedPublication	PreviewPipeline	Preview	XfmdDesktop	f0721
PagedPublication	RenderLayout	Renderer	XfmdDesktop	f0841
PagedPublication	SourceAnchorMapping	Viewport	XfmdDesktop	f0912
PagedPublication	TypographyAndDrawing	Typography	XfmdDesktop	f0949
PagedPublication	WorkspaceControls	NativeShell	XfmdDesktop	f1019
SynchronizedScrolling	FoxPresentation	NativeShell	XfmdDesktop	f0358

SynchronizedScrolling	RenderLayout	Renderer	XfmdDesktop	f0842
SynchronizedScrolling	ScrollMotion	Viewport	XfmdDesktop	f0891
SynchronizedScrolling	SourceAnchorMapping	Viewport	XfmdDesktop	f0913
SynchronizedScrolling	WorkspaceControls	NativeShell	XfmdDesktop	f1020
WorkPath Navigation	FilteredFileTree	Workspace	XfmdDesktop	f0350
WorkPath Navigation	WorkPathHistory	Workspace	XfmdDesktop	f0997
WorkPath Navigation	WorkspaceControls	NativeShell	XfmdDesktop	f1022